Interpretive Caption: Real-Time Vocal Emotion Cues for DHH Users

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Abstract

Deaf and Hard-of-Hearing (DHH) individuals increasingly rely on real-time captioning to access spoken content in educational and professional settings. However, traditional captions omit vocal emotional cues, such as intonation and affect which can hinder comprehension and engagement. This work introduces *Interpretive Caption*, a machine-learning prototype that augments captions with emotion-aware annotations derived from vocal tone. Using letter-coded tags with hover-based tooltips, the system conveys emotional context on demand, balancing clarity with cognitive accessibility. We conducted a qualitative study with eight DHH participants who interacted with the prototype and shared feedback on usability, emotional clarity, and layout design. Findings highlight the value of hover-based emotional cues, customization features, and segmentation aligned with cognitive load principles. Participants appreciated the non-intrusive emotional insights, while also identifying areas for improvement, including accent-inclusive emotion recognition and better mobile accessibility. Our contributions include a real-time captioning prototype integrating speech emotion recognition, a user-controllable emotion display interface, and design insights for affective accessibility in educational contexts. This work offers a foundation for inclusive, expressive captioning and informs future multimodal caption systems that prioritize interpretability, cultural sensitivity, and user agency.

CCS Concepts

• **Human-centered computing** → **Accessibility**; **Accessibility systems and tools**; **Accessibility design and evaluation methods**;

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Keywords

Accessibility, Real-time Captioning, Emotional Expression, Deaf and Hard-of-Hearing

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1 Introduction & Related Work

Captioning technology has significantly improved communication access for Deaf and Hard-of-Hearing (DHH) individuals in classrooms, video conferencing, and online platforms [16], enabling users to follow spoken content through text.

In STEM classrooms, however, captioning alone may be insufficient. Instructors often use vocal emphasis, pitch, and affect to clarify complex concepts [20, 33], but ASR systems typically omit these non-verbal cues, limiting comprehension. While recent affective captioning systems [15, 39] integrate speech emotion recognition and large language models, most target entertainment [19, 21] rather than educational contexts, which pose distinct challenges like rapid topic shifts [17].

According to Dual Coding Theory [7], combining verbal and non-verbal information enhances cognitive processing. Without emotional context, caption users miss out on non-verbal cues, and may face increased cognitive load [24]. Prior emotive captioning research [2, 4, 22, 23, 29] has explored ways to integrate vocal tone, facial expressions, and body language to improve DHH users' experiences. These cues are vital to communication [8, 12, 28, 34], but methods that rely on color codes or stylized prosody often require memorization, which can be cognitively taxing. Additionally, the split-attention effect [14, 25]—where users divide focus between the speaker and caption display—may increase processing difficulty. Systems like Caption Royale [5] used font styling to encode emotion, but required users to recall formatting combinations. Our approach addresses this by using minimal, hover-based pop-ups

that reveal emotional tone only when needed, reducing clutter and effort.

We present **Interpretive Caption**, a real-time captioning prototype that embeds vocal intonation into real-time ASR transcription. It combines OpenAI Whisper for transcription [31] with a fine-tuned Wav2Vec2 model trained on CREMA-D and RAVDESS datasets [10] for emotion recognition. Detected emotions appear as single-letter tags with hover-based tooltips [36, 38], avoiding the need for memorizing emotionally-coded visuals—like emoji sets or color combinations—commonly used in affective captioning systems [2, 5, 6].

Unlike other systems that explore facial expressions [13, 37], dynamic subtitles [3], or overlays [18, 32, 35], our contribution focuses solely on vocal intonation. By offering letter-coded emotional cues with corresponding hover explanations, we aim to provide a lightweight, cognitively accessible captioning experience tailored for DHH students in live STEM instruction setting. A limitation is that we have not yet conducted formal cognitive load evaluations (e.g., NASA-TLX, EEG), which is a direction for future work.

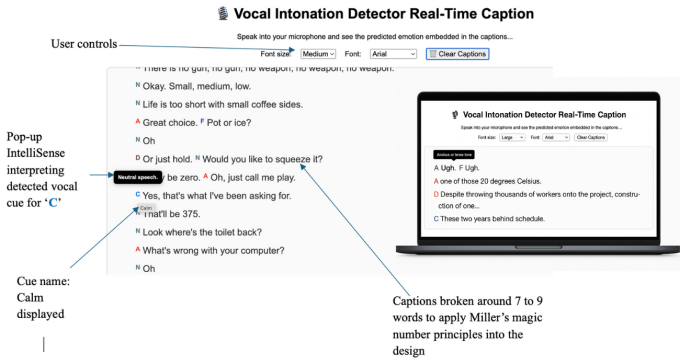


Figure 1: Vocal Intonation Detector Real-Time Caption system. The interface presents real-time captions with embedded letter-coded vocal tone tags (e.g., "F" for fearful, "A" for angry), enhanced by IntelliSense-style tooltips showing corresponding tone descriptions such as "Anxious or tense tone" or "Calm." Captions are segmented into reduced word chunks following Miller's magic number principle to aid comprehension and reduce cognitive load. Segmenting into reduced word chunks reflects Miller's principle to optimize short-term memory load; while some users noted choppiness, this approach supports chunking and will be compared with alternative strategies in future studies.

2 Interpretive Caption Prototype Design and Rationale

Interpretive Caption integrates two pretrained models—OpenAI Whisper for real-time transcription and Wav2Vec2-Large-XLSR for speech emotion recognition—to enrich captions with vocal intonation. The system processes audio input through Whisper for ASR and Wav2Vec2 for emotion tagging, as shown in Figure 2. Figure 1 shows the customizable web interface. The interface allows users to adjust font size and style, and includes a clear-caption function.

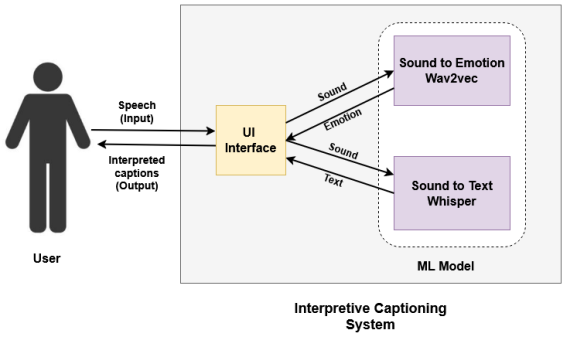


Figure 2: The architecture of the Interpreting Captioning System showing the interactions between the users and system and the data processing within the system.

We segmented the captions into reduced word chunks based on Miller's "magic number" [26] to support readability and cognitive processing. Each line includes a letter-coded emotion tag (e.g., "H" for Happy) with hoverable tooltips (e.g., "Cheerful tone"). Each caption segment is paired with a corresponding detected emotion based on pitch contour, vocal intensity, and prosodic rhythm, using thresholds trained on datasets like CREMA-D and RAVDESS. The selected emotion is then converted to a one-letter tag (e.g., 'A' for Angry), which the user can interpret by hovering to reveal the descriptive tooltip. While aligned with cognitive theory, we acknowledge the absence of formal cognitive load metrics—a limitation of this study.

Emotion classification uses a Wav2Vec2 model fine-tuned on datasets like CREMA-D, RAVDESS, TESS, and SAVEE [11], detecting emotions such as calm, excited, sad, and angry at 73% accuracy [9]. Whisper achieves a 5.5% word error rate (WER) on the LibriSpeech benchmark [31]. Synchronized caption and emotion output is delivered via Flask-SocketIO and WebSockets. Table 1 lists the emotion labels.

Technical Validation. The Whisper model, trained on approximately 680,000 hours of web audio, achieves a word error rate (WER) of around 5.5% on established benchmarks [31]. Similarly, the Wav2Vec2-Large-XLSR model reaches 73% accuracy in emotion recognition tasks [9], supporting its use for reliable real-time transcription and affect detection.

Design Rationale. This prototype addresses expressive gaps in traditional captions—especially for DHH students in STEM settings—where vocal tone often conveys emphasis and meaning [1, 20]. It is grounded in Dual Coding Theory [7], and the design integrates verbal content with non-verbal vocal cues to support comprehension and reduce cognitive load. More so, unlike icon- or color-coded methods that can require prior training or increase screen clutter [6, 32], our approach uses compact, single-letter emotion tags paired with hover-based tooltips. This allows users to access affective context only when needed, promoting user control and minimizing distraction. While letter abbreviations (e.g., 'S' for Sad) still involve a learning curve, they are paired with hover-tooltips and do not compete for visual space like emojis or color

Table 1: Vocal intonation labels and descriptions.

Emotion	Abbr.	Description
Angry	A	Raised or frustrated voice
Calm	C	Relaxed or neutral speech
Disgust	D	Tone showing discomfort or disapproval
Fearful	F	Anxious or tense vocalization
Happy	H	Cheerful or joyful tone
Neutral	N	Flat or neutral tone
Sad	S	Low energy voice
Surprised	U	Tone of unexpected reaction
Excited	E	High-energy, engaged tone
Forceful	Fo	Assertive or commanding tone

codes, which participants in prior work [5] found distracting or confusing.

Current limitations include the absence of facial and body language cues, and reduced emotion classification accuracy in noisy environments. Future iterations aim to incorporate broader multimodal inputs for a more comprehensive interpretive experience.

3 Study Design

To evaluate the usability and effectiveness of the Interpretive Caption from user perspective, we conducted a qualitative study [30] with eight DHH participants recruited from academic and community networks. Most identified as students (75%) and educators (25%), with a range of hearing statuses, including DHH, and one cognitive hearing-related condition. Caption usage varied, with 63% using them daily. While five minutes of interaction provided initial feedback, we acknowledge that longitudinal evaluation is needed to assess long-term usability and learning impact.

Each 30-minute remote session included a walkthrough of the prototype's features—such as font customization and hover-based emotion cues—followed by an interactive exploration of emotion-tagged captions within a simulated lecture scenario. Participants then engaged in a semi-structured interview focused on usability, emotional comprehension, layout preferences, and feature suggestions.

Interview questions addressed captioning habits, clarity of emotional cues, system intuitiveness, and desired improvements. We used Braun and Clarke's six-phase thematic analysis [27]. Two researchers independently coded transcripts using NVivo and resolved discrepancies through discussion.

This study offers early-stage insights into affect-aware captioning design for DHH users in educational settings and lays the groundwork for future longitudinal and comparative evaluations.

4 Results and Discussion

Our qualitative study with eight DHH participants highlighted the promise of the Interpretive Caption prototype in improving emotional comprehension, personalization, and accessibility in STEM learning. These insights both align with and expand upon prior work in affective captioning and accessible communication.

4.1 Emotional Awareness Enhances Comprehension and Connection

Participants consistently emphasized that emotional cues deepened their understanding of speaker intent, reinforcing the communicative value of vocal tone in educational discourse. This supports findings by Rashid et al.[39], who demonstrated that affective cues improve engagement and comprehension. However, unlike prior systems that rely on visual clutter (e.g., emojis or color codes), users valued our minimalist design that deferred interpretation via hover—though not all participants used the hover feature consistently, several appreciated that emotional information wasn't persistently visible, reducing visual distraction. This aligns with Dual Coding Theory [7], suggesting that combining verbal and nonverbal inputs improves cognitive processing. Several participants noted that even subtle emotional context—such as distinguishing between calm and frustrated tones—reduced ambiguity in technical real-time conversations, an extension of what Marschark et al. [20] described as "intention gap" in DHH educational access.

4.2 On-Demand Cues Support Cognitive Efficiency

The system's hover-based design balanced expressivity and visual clarity. This contrasts with prior work like Caption Royale [5], which required users to memorize color and font combinations, increasing cognitive load. Participants appreciated our tooltips for their discoverability and voluntary engagement, which mirrors findings by Viegas et al. [37] that interactivity can mitigate information overload when implemented judiciously. Further, users' emphasis on customization aligns with literature on adaptive captioning interfaces [29], which promote agency and reduce cognitive fatigue. Some participants requested clearer onboarding and platform compatibility—echoing similar concerns raised in dynamic subtitle systems like Bechtold et al. [3], where non-desktop interactions were underserved.

4.3 Segmentation: Supporting or Hindering Flow?

While our use of Miller's "magic number"[25] in split-attention scenarios is not a conventional captioning standard, participants noted that chunked segments aided readability, though some found them choppy. While this suggests cognitive benefits, formal validation (e.g., NASA-TLX) is needed to confirm these effects. This suggests that while segmentation may support load management in theory, its implementation must be carefully calibrated to maintain narrative coherence in real-time flow.

4.4 Accent Sensitivity and Inclusive Modeling

Participants noted that emotional accuracy dropped with accented speech, highlighting a challenge in speech emotion recognition that has also been acknowledged by Noroozi et al.[35]. Our findings reinforce the need for culturally and linguistically inclusive datasets and the incorporation of speaker diarization in real-time systems [13]. This concern is particularly relevant as affective computing research increasingly recognizes cultural variability in emotional expression and interpretation [34]. As one participant put it, "The emotion

tags were helpful, but sometimes they felt wrong—maybe because of my accent?" Future models must be trained on globally diverse data to ensure equitable access.

4.5 Design Contributions in Context

Compared to prior emotive captioning interfaces that use static visuals [2, 22], our prototype advances the field by:

- Introducing letter-coded, hover-based emotional tags that reduce visual clutter and enhance optionality.
- Our segmentation strategy aligns with cognitive theory (e.g., Miller's law) and is refined by real user feedback, supporting both cognitive efficiency and user preference.
- Prioritizing inclusive emotion recognition and on-demand emotional interpretation rather than requiring memorization or passive consumption of stylized elements. For instance, users who reported difficulty distinguishing between neutral and sad tones suggested color-descriptive tooltips or optional iconography. As a result, we propose future iterations include multi-modal cues (e.g., haptics, icons) with user toggles for emotional granularity. This positions Interpretive Caption as a user-controllable, cognitively considerate system that serves both educational and accessibility goals.

5 Conclusion and Future Work

This study introduced Interpretive Caption, a machine-learning-based system that enriches real-time captions with vocal intonation cues to support expressive, accessible communication for DHH users. A qualitative evaluation with eight participants in STEM contexts revealed both the potential and challenges of emotion-aware captioning, including the need for clearer visual cues, speaker identification, and culturally inclusive emotion recognition. Findings underscore the importance of context-aware design and user control to balance expressivity with cognitive load.

Future work will involve longitudinal studies to examine sustained user interaction and adaptation, along with broader comparisons across captioning strategies—such as emotional layering, color-coded, and emoji-based systems—to inform inclusive design guidelines. We also aim to adapt models for STEM discourse, expand cultural and linguistic representation, test mobile responsiveness, and formally evaluate cognitive load using tools like NASA-TLX.

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